

exist some t_1 with $0 < t_1 < t < t_0$ such that $d_X(o, \gamma(t_1)) \leq R$, contradicting the definition of t_0 . Thus, $\gamma(t_0)$ is an accumulation point of $\text{Cone}_o(U)$ and $\gamma(t_0) \in S_X(o, R)$.

Suppose p is an accumulation point of $\text{Cone}_o(U)$ and $p \in S_X(o, R)$. Then there exists a sequence $p_n \in \text{Cone}_o(U)$ such that $p_n \rightarrow p$ as $n \rightarrow \infty$. Let $x_n \in [o, p_n] \cap U$ for some geodesic segment joining o to p_n .

$$\begin{aligned} d_X(x_n, p) &\leq d_X(x_n, p_n) + d_X(p_n, p) \\ &= d_X(o, p_n) - d_X(o, x_n) + d_X(p_n, p) \\ &= d_X(o, p_n) - R + d_X(p_n, p) \xrightarrow{n \rightarrow \infty} d_X(o, p) - R + 0 = 0. \end{aligned}$$

Thus $x_n \rightarrow p$ as $n \rightarrow \infty$. Since U is a connected component of S , it is closed, and we have $p \in U$. From this above discussion, we get $\gamma(t_0) \in U$.

Define $t_1 := \sup\{t \geq 0 \mid d_X(o, \gamma(t)) \leq R\}$, then similarly we can argue that $\gamma(t_1) \in V$.

Note that

$$\begin{aligned} d_X(\gamma_1(t), \gamma_2(t)) &= d_X(\gamma_1(t), \gamma(t_0)) + d_X(\gamma(t_0), \gamma(t_1)) + d_X(\gamma(t_1), \gamma_2(t)), \\ d_X(o, \gamma_1(t)) &= d_X(o, \gamma_1(R)) + d_X(\gamma_1(R), \gamma_1(t)) = R + d_X(\gamma_1(R), \gamma_1(t)), \\ d_X(o, \gamma_2(t)) &= d_X(o, \gamma_2(R)) + d_X(\gamma_2(R), \gamma_2(t)) = R + d_X(\gamma_2(R), \gamma_2(t)), \end{aligned}$$

which gives,

$$\begin{aligned} 2(\gamma_1(t)|\gamma_2(t))_o &= 2R - d_X(\gamma(t_0), \gamma(t_1)) \\ &\quad + d_X(\gamma_1(t), \gamma(t_0)) - d_X(\gamma_1(R), \gamma_1(t)) \\ &\quad + d_X(\gamma(t_1), \gamma_2(t)) - d_X(\gamma_2(R), \gamma_2(t)). \end{aligned} \tag{42}$$

For $x \in U$ and $y \in V$ we have $2(x|y)_o = d_X(x, o) + d_X(y, o) - d_X(x, y) = 2R - d_X(x, y)$. Thus from (42) and application of the triangle inequality we get,

$$\begin{aligned} 2|(\gamma_1(t)|\gamma_2(t))_o - (x|y)_o| &\leq |d_X(x, y) - d_X(\gamma(t_0), \gamma(t_1))| \\ &\quad + |d_X(\gamma_1(t), \gamma(t_0)) - d_X(\gamma_1(R), \gamma_1(t))| \\ &\quad + |d_X(\gamma(t_1), \gamma_2(t)) - d_X(\gamma_2(R), \gamma_2(t))| \\ &\leq d_X(x, \gamma(t_0)) + d_X(\gamma(t_1), y) + d_X(\gamma(t_0), \gamma_1(R)) + d_X(\gamma(t_1), \gamma_2(R)) \\ &\leq 2(\text{diam}(U) + \text{diam}(V)) \end{aligned} \tag{43}$$

We know from property of Gromov product space

$$\lim_{t \rightarrow \infty} (\gamma_1(t)|\gamma_2(t))_o = (\xi|\eta)_o.$$

Therefore, from (43) we can conclude the desired claim (40) (by taking limit $t \rightarrow \infty$ on the left hand side).

For proving (41), consider geodesic rays γ_1 and γ_2 joining o to ξ and η , respectively, as above. Then $(\gamma_1(t)|\gamma_2(t))_o \uparrow (\xi|\eta)_o$ as $t \rightarrow \infty$. Note $\gamma_1(R), \gamma_2(R) \in U$. Thus,

$$(\xi|\eta)_o \geq (\gamma_1(R)|\gamma_2(R))_o = \frac{1}{2} \left(d_X(o, \gamma_1(R)) + d_X(o, \gamma_2(R)) - d_X(\gamma_1(R), \gamma_2(R)) \right) \geq R - \frac{\text{diam}(U)}{2}.$$

□

Now we are well equipped to give the proof of **Theorem 1.5**.

Proof of Theorem 1.5. Since (X_n, x_n) is a maximal Gromov product space, it is isometric to $\mathcal{M}(\partial_P X_n, \rho_{x_n})$, for each n , via the visual embedding (see **Proposition 2.2.6**). Similarly, (X, x) is isometric to $\mathcal{M}(\partial_P X, \rho_x)$. Define the sequence of antipodal spaces $(Z_n, \rho_n) = (\partial_P X_n, \rho_{x_n})$ for all n and $(Z, \rho_0) = (\partial_P X, \rho_x)$, where Z is a finite set of cardinality $m < \infty$.

By the given hypothesis the Moebius spaces $(\mathcal{M}(Z_n), \rho_n) \xrightarrow{GH \text{ conv.}} (\mathcal{M}(Z), \rho_0)$. We will demonstrate that for any given $\delta > 0$, it is possible to construct δ -isometries $f_n: (Z_n, \rho_n) \rightarrow (Z, \rho_0)$ for all sufficiently large n . Thus, by **Remark 3.5**, this will be sufficient to conclude $(Z_n, \rho_n) \xrightarrow{AI \text{ conv.}} (Z, \rho_0)$, i.e. $(\partial_P X_n, \rho_{x_n}) \xrightarrow{AI \text{ conv.}} (\partial_P X, \rho_x)$.

Since $(\mathcal{M}(Z_n), \rho_n) \xrightarrow{GH \text{ conv.}} (\mathcal{M}(Z), \rho_0)$, for every $R > 0$, there exists ϵ_n -isometries

$$\mathcal{F}_n: B_{\mathcal{M}(Z)}(\rho_0, R) \rightarrow B_{\mathcal{M}(Z_n)}(\rho_n, R)$$

such that $\epsilon_n \rightarrow 0+$ and $\mathcal{F}_n(\rho_0) = \rho_n$.

The antipodal space (Z, ρ_0) is of finite cardinality, let us label $Z = \{1, 2, \dots, m\}$. By [BP24, Theorem 1.1], the maximal Gromov hyperbolic spaces $\mathcal{M}(Z, \rho_0)$ is isometric to a polyhedral complex such that